```cpp
#include <iostream>
using namespace std;
int steps = 0;
int fibRecursive(int n) {
    steps++;
    if (n <= 1) return n;
    return fibRecursive(n - 1) + fibRecursive(n - 2);
}
int main() {
    int n;
    cout << "Enter n: ";
    cin >> n;
    cout << "\nIterative Fibonacci\n";
    int a = 0, b = 1, c;
    int count = 0;
    cout << "Initially: a = " << a << ", b = " << b << endl;
    if (n <= 1) {
        cout << "n is 0 or 1, so Fib(" << n << ") = " << n << endl;
        c = n;
    } else {
        for (int i = 2; i <= n; i++) {
            c = a + b;
            count++;
            cout << "Step " << count << " (i = " << i << "): c = a + b = " << a << " + " << b << " = "
                    << c << endl;
            a = b;
            b = c;
            cout << "         update: a = " << a << ", b = " << b << endl;
        }
    }
    cout << "Iterative: Fib(" << n << ") = " << c << ", steps = " << count << endl;
    steps = 0;
    int r = fibRecursive(n);
    cout << "\nRecursive: Fib(" << n << ") = " << r << ", steps = " << steps << endl;
    return 0;
}
```

```
PS C:\Users\fishb\Desktop\Sem 7\DAA> cd "c:\Users\fishb\Desktop\Sem 7\DAA\" ; if ($?) { g++ 1_fibonacci.cpp -o 1_fibonacc
i } ; if ($?) { .\1_fibonacci }
Enter n: 8

Iterative Fibonacci
Initially: a = 0, b = 1
Step 1 (i = 2): c = a + b = 0 + 1 = 1
         update: a = 1, b = 1
Step 2 (i = 3): c = a + b = 1 + 1 = 2
         update: a = 1, b = 2
Step 3 (i = 4): c = a + b = 1 + 2 = 3
         update: a = 2, b = 3
Step 4 (i = 5): c = a + b = 2 + 3 = 5
         update: a = 3, b = 5
Step 5 (i = 6): c = a + b = 3 + 5 = 8
         update: a = 5, b = 8
Step 6 (i = 7): c = a + b = 5 + 8 = 13
         update: a = 8, b = 13
Step 7 (i = 8): c = a + b = 8 + 13 = 21
         update: a = 13, b = 21
Iterative: Fib(8) = 21, steps = 7

Recursive: Fib(8) = 21, steps = 67
PS C:\Users\fishb\Desktop\Sem 7\DAA>
```